

TABULA-R²: A Unified Benchmark and Toolkit for Table Reasoning

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<https://github.com/fyq5166/TABULA-R2>

Overview. This short document summarizes the motivation, methodology, key findings, and significance of my recent work on **TABULA-R²**, a benchmark and open-source toolkit designed to make large-language-model (LLM) evaluation on tabular reasoning both reproducible and interpretable. Existing table-reasoning evaluations often mix free-form prompts, inconsistent metrics, and untraceable validation. TABULA-R² addresses this gap by introducing a layered framework that unifies natural-language prompting, symbolic execution, and automatic validation under a common protocol.

What I built. The project began with the design and construction of the **TABULA-R² benchmark**, a curated corpus of **129 real-world data tables** derived from *Our World in Data* (OWID) and related repositories. Each table was normalized into a consistent schema (CSV + metadata) and annotated with domain tags spanning **8 thematic areas**—health, demographics, economics, energy, education, climate, environment, and policy. On top of these tables, I authored and verified **252 question–answer pairs** that test diverse reasoning patterns:

- **Arithmetic:** quantitative aggregation and comparison;
- **Conditional:** logical filtering and multi-condition inference;
- **Proxy inference:** reasoning via indirect indicators;
- **Entity alignment:** relational joins across multiple tables.

Each question is labeled for complexity, answerability, reasoning type, and domain, producing a fine-grained analytical backbone for controlled evaluation.

Building on this dataset, I implemented a fully integrated **execution system** that operationalizes the entire reasoning loop through a structured **PLAN/END protocol**. The pipeline connects natural-language input to symbolic execution and validation through the following components:

- A compact **domain-specific language (DSL)** supporting operations such as `filter_all`, `groupby().agg()`, and `derive()`, designed for deterministic replay and cross-model comparability.
- A modular **validator–executor** that scores both end answers and every intermediate DSL step ($\approx 3,000+$ step-level records across runs).
- Automated logging, visualization, and statistical modules for success-rate analysis, per-step diagnostics, and error-type clustering.

To support reproducibility, I developed a comprehensive **documentation and configuration layer**: each experiment is encapsulated in YAML-style configs under `configs/`, with 9 baseline variants and 50 comparison subsets; corresponding analyses are scripted in `analysis/` and visualized under `data/figures/`. A **technical report** (86 pages) documents the full pipeline, metrics, and findings, including controlled comparisons across models (llama3, llama3.2, gemma, qwen, codellama), few-shot variants ($k=0,1,3$), CoT inclusion, and observation modes (`header_1`, `header_5`, `full_table`).

All code, data, and reproducible experiment logs are openly available at:
<https://github.com/fyq5166/TABULA-R2>.

Study design. Using this unified pipeline, I conducted three large-scale analyses:

1. **Single vs. Multi-Table Baselines** — examining the degradation in reasoning stability when relational joins are required.
2. **Distractor Effects** — injecting additional tables (relevant, irrelevant, misleading, none) to quantify how noise and context influence behavior.
3. **Controlled Comparisons** — systematically varying model family, few-shot exemplars, chain-of-thought (CoT) inclusion, and observation scope to isolate protocol sensitivity.

Key findings.

- *Multi-table reasoning remains fragile*: success rate drops by half compared with single-table settings, and DSL step reliability falls sharply.
- *Distractors can help, not only hurt*: relevant contextual tables improve schema anchoring, whereas misleading ones trigger over-fitting.
- *Protocol alignment dominates model scale*: zero-shot prompting with concise schema exposure (`header_1`) outperforms few-shot or CoT setups under strict PLAN/END grammar.
- *Proxy and entity-alignment reasoning* persist as the weakest categories, suggesting a need for stronger symbolic grounding.

Significance. TABULA-R² offers a process-aware, statistically grounded diagnostic lens for LLM reasoning on structured data. By combining executable DSLs with standardized validation, it bridges the gap between natural-language QA and formal logic evaluation—allowing finer-grained error attribution and model comparison. The framework emphasizes *reproducibility*, *interpretability*, and *protocol transparency*, providing a template for future open-source reasoning benchmarks.

Limitations. The current validator remains tightly coupled to the PLAN/END grammar, limiting cross-model generality. Sample sizes in certain reasoning categories are small, widening confidence intervals. Moreover, observation-design choices (`header_k` vs. full table) can bias schema visibility and affect outcomes. Addressing these constraints requires larger-scale runs, per-model format calibration, and protocol-agnostic normalization.

Future directions. I plan to (i) extend the DSL to support compositional joins and natural-language control flow, (ii) integrate AST-level matching for validator robustness, (iii) scale

cross-model comparisons with SOTA LLMs under identical symbolic settings, and (iv) release annotated human-audit subsets for qualitative reasoning analysis.

Full codebase, datasets, and reproducible experiment logs are available at:

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